Sudharshan Ravi

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Summary

- **My Profile:** Computational Researcher at the University of Lausanne (UNIL) | Ph.D. in Chemical and Bioengineering (Focus on Computational Systems Biology) at ETH Zürich | Swiss resident (valid work permit) | Indian citizen | 10+ years in Bioinformatics Research | 7+ years in Metabolic Network Analysis | 5+ years as Lead Researcher
- **My work:** Integrative high-dimensional multi-omics data analysis in understanding the biology of aging and age-related diseases | Novel workflow for the analysis and integration of omics data in genome-scale metabolic models | Inferential analysis on the impact of socio-economic and other exogenous gradients on transcriptomic patterns and health outlook

Experience

Premier Assistant — *Lausanne, Switzerland*

DEPARTMENT OF FUNDAMENTAL MICROBIOLOGY (DMF), UNIVERSITY OF LAUSANNE (UNIL) — *Mar. 2026 – Present*

- Developing constraint-based computational models to characterize metabolic interactions and resource exchange networks in microbial communities, advancing mechanistic understanding of community-level metabolic dynamics.

Bioinformatics Lead — *London, UK*

MYOLYTICS — *Jan. 2025 – Feb. 2026*

- Tailored machine learning models to longitudinal multimodal data, integrating molecular and biosensor measurements, for biomarker discovery and pre-symptomatic prediction of sports-related muscle injuries in professional athletes.
- Translated predictive insights into personalized injury mitigation, management and recovery strategies, supporting athletes in maximizing performance and reducing time away from competition.

Postdoctoral Research Associate — *Zürich, Switzerland*

SOCIAL GENOMICS, JACOBS CENTER FOR PRODUCTIVE YOUTH DEVELOPMENT, UZH ZÜRICH — *Mar. 2021 – Jan. 2025*

- Characterized the molecular consequences of socioeconomic inequalities across transcriptomic, immunological and metabolic axes using integrative multi-omics and network-based approaches on a longitudinal cohort spanning adolescence to young adulthood.
- Revealed disruption of upstream immune regulatory networks as a key mechanism connecting early-life socioeconomic disparities to altered metabolic function, accelerated aging and increased disease predisposition.

Graduate Research Assistant / Doctoral Candidate — *Zürich, Switzerland*

CHEMICAL AND BIOLOGICAL SYSTEM ENGINEERING LABORATORY, ETH ZÜRICH — *Jan. 2016 – Sep. 2020*

- Bioinformatic analysis of human aging: Extensive analysis of population-level transcriptomic data characterizing the biology of aging and its close relationship to metabolism.
- ΔFBA: Developed a novel pipeline integrating differential omics into genome-scale metabolic models using constraint-based modeling and mathematical optimization.
- Multi-omics analyses: Deciphered the chromatin – gene expression interplay in cell fate commitment of hematopoietic cells. Illuminated the mechanisms of constrained antitumor immune responses during IDO1 inhibition of ovarian cancers.

Education

Ph.D. in Chemical and Bioengineering *Zürich, Switzerland*

ETH Zürich | Visiting researcher at University at Buffalo, USA *(Aug. 2018 – Jul. 2020)* *Jan. 2016 – Sep. 2020*

- Thesis: Metabolic network analysis and its application in understanding the biology of aging – Prof. Dr. Rudiyanto Gunawan
- Research Specialization: Bioinformatics | Systems Biology | Aging & Age-Related Diseases | Metabolic Network Analysis | Constraint-Based Modeling | Multi-omics Analysis

M.Sc. in Chemical and Bioengineering *Zürich, Switzerland*

ETH Zürich *Sep. 2013 – Sep. 2015*

- Thesis: Reducing Complexity — Efficient Modeling for Biorefinery Concepts – Prof. Dr. Konrad Hungerbühler
- Research Specialization: Surrogate Modeling | Sustainable Engineering | Machine Learning | Bioprocessing Technologies | Process Engineering

B.Tech. in Industrial Biotechnology *Thanjavur, India*

SASTRA University | Harvard – MIT Health Sciences and Technology *Sep. 2008 – Oct. 2012*

- Thesis: An Induced CD44v6 – EGFR scaffold kinase interaction predisposes a subpopulation of chemotherapy-tolerant breast cancer cells and serves as a bivalent target of resistance – Prof. Dr. Aaron Goldman
- Research Specialization: Molecular Biology | Genetics | Cancer Biology | Drug Resistance | Nanomedicine

Selected Publications

[1] **Sudharshan Ravi**, Michael J. Shanahan, Brandt Levitt, *et al.* Socioeconomic inequalities in early adulthood disrupt the immune transcriptomic landscape via upstream regulators. **Scientific Reports**, 2024

[2] **Sudharshan Ravi** and Rudiyanto Gunawan. ΔFBA – Predicting metabolic flux alterations using genome-scale metabolic models and differential transcriptomics data. **PLOS Computational Biology**, 2021

[3] Emelyne Teo, **Sudharshan Ravi**, Diego Barardo, *et al.* Metabolic stress is a primary pathogenic event in transgenic *Caenorhabditis elegans* expressing pan-neuronal human amyloid beta. **eLife**, 2019

See complete list of publications and conference presentations at **Google Scholar**.

Proficiencies

Bioinformatics	R | Bioconductor | MATLAB | COBRA Toolbox | Python
Infrastructure	Bash | Git | Docker | High-Performance Computing
Networks	Cytoscape | STRING | iGraph | Gephi
Visualization	ggplot2 | Adobe Illustrator | Inkscape | PowerPoint | LaTeX
Laboratory	Cell culture | IHC | Western Blotting | Flow cytometry | Fluorescent microscopy
Languages	English (Fluent) | German (Intermediate) | Tamil (Native) | Hindi (Beginner)

Academic Merits & Activities

Scholarships

2015	Birkigt Scholarship Award, ETH Zürich
2012	Undergraduate Research Scholarship, SASTRA University & Harvard – MIT HST

Mentoring

2016 – 2018	Tutored Bioprocess Modeling, Process Simulation and Flowsheeting and Statistical and Numerical Methods courses at ETH Zürich
2016 – 2019	Supervised 2 MS students in research projects and semester theses at ETH Zürich & University at Buffalo